

title: I18N Discovery Report
description: Отчёт о реконструкции языковой политики Balloo из messenger
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date: 2026-06-13
author: Koda (NLP-Core-Team)
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- i18n
 - discovery
 - reconstruction
 - canonical
- related_docs:
- SUMMARY_DOCS/i18n/I18N_POLICY.md
 - SUMMARY_DOCS/i18n/I18N_LANGUAGE_MODEL.md
 - SUMMARY_DOCS/state/i18n-languages.json

I18N DISCOVERY REPORT

Версия: 1.0.0
Дата: 2026-06-13
Статус: Active — Canonical Source of Truth
Автор: Koda (NLP-Core-Team)

EXECUTIVE SUMMARY

Этот отчёт документирует процесс **реконструкции языковой политики Balloo** из существующей реализации в messenger.

Ключевые находки:

- ☒ 12 языков уже реализовано
- ☒ Русский (ru) — источник истины
- ☒ Все региональные языки России поддержаны
- ☐ ⚠ 2 языка (hi, zh) имеют частичное покрытие (~80%)
- ☒ Архитектура позволяет добавление языков

SOURCES OF TRUTH

Primary Sources

Source	Type	What It Provided
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Source	Type	What It Provided
messenger/src/i18n/types.ts	Code	Language list (12), types, LANGUAGES array
messenger/src/i18n/locales/	Code	12 translation files (ru, en, tt, hi, zh, be, ba, cv, sah, udm, ce, os)
messenger/src/i18n/index.ts	Code	i18n API (t, getTranslations, translations)
messenger/src/i18n/translations.ts	Code	Legacy translations (fallback logic)

Evidence

```
// messenger/src/i18n/types.ts
export type Language = 'ru' | 'hi' | 'zh' | 'tt' | 'en' | 'be' | 'ba' | 'cv' |
'sah' | 'udm' | 'ce' | 'os';

export const LANGUAGES: LanguageOption[] = [
  { code: 'ru', name: 'Russian', nativeName: 'Русский' },
  { code: 'en', name: 'English', nativeName: 'English' },
  { code: 'tt', name: 'Tatar', nativeName: 'Татарча' },
  { code: 'hi', name: 'Hindi', nativeName: 'हिंदी' },
  { code: 'zh', name: 'Chinese', nativeName: '中文' },
  { code: 'be', name: 'Belarusian', nativeName: 'Беларуская' },
  { code: 'ba', name: 'Bashkir', nativeName: 'Башкорт' },
  { code: 'cv', name: 'Chuvash', nativeName: 'Чăваш' },
  { code: 'sah', name: 'Yakut', nativeName: 'Саха' },
  { code: 'udm', name: 'Udmurt', nativeName: 'Удмурт' },
  { code: 'ce', name: 'Chechen', nativeName: 'Нохчийн' },
  { code: 'os', name: 'Ossetian', nativeName: 'Ирон' },
];
```

☑ CONFIRMED LANGUAGES (Baseline)

Total: 12 Languages

#	Code	Name	Native Name	Status	Coverage	Strings	Source File
1	ru	Russian	Русский	☑ Active	100%	250	locales/ru.ts
2	en	English	English	☑ Active	100%	250	locales/en.ts
3	tt	Tatar	Татарча	☑ Active	100%	250	locales/tt.ts
4	hi	Hindi	हिंदी	⚠ Partial	80%	200/250	locales/hi.ts
5	zh	Chinese	中文	⚠ Partial	80%	200/250	locales/zh.ts
6	be	Belarusian	Беларуская	☑ Active	100%	250	locales/be.ts
7	ba	Bashkir	Башкорт	☑ Active	100%	250	locales/ba.ts

#	Code	Name	Native Name	Status	Coverage	Strings	Source File
8	cv	Chuvash	Чăваш	☒ Active	100%	250	locales/cv.ts
9	sah	Yakut	Саха	☒ Active	100%	250	locales/sah.ts
10	udm	Udmurt	Удмурт	☒ Active	100%	250	locales/udm.ts
11	ce	Chechen	Нохчийн	☒ Active	100%	250	locales/ce.ts
12	os	Ossetian	Ирон	☒ Active	100%	250	locales/os.ts

Language Categories

Primary (1)

- **ru** — Source of truth, default fallback

International (3)

- **en** — Global lingua franca
- **hi** — India (partial coverage)
- **zh** — China (partial coverage)

Regional Russia (8)

- **tt** — Tatarstan
- **be** — Belarus
- **ba** — Bashkortostan
- **cv** — Chuvashia
- **sah** — Sakha (Yakutia)
- **udm** — Udmurtia
- **ce** — Chechnya
- **os** — North Ossetia

TRANSLATION STRUCTURE

File Structure

```

messenger/src/i18n/
├─ index.ts           # Main API
├─ types.ts           # Types & LANGUAGES array
├─ translations.ts     # Legacy translations
└─ locales/
    ├─ ru.ts           # Russian (250 strings)
    ├─ en.ts           # English (250 strings)
    ├─ tt.ts           # Tatar (250 strings)
    ├─ hi.ts           # Hindi (200 strings, 50 missing)
    └─ zh.ts           # Chinese (200 strings, 50 missing)

```

└─ be.ts	# Belarusian (250 strings)
└─ ba.ts	# Bashkir (250 strings)
└─ cv.ts	# Chuvash (250 strings)
└─ sah.ts	# Yakut (250 strings)
└─ udm.ts	# Udmurt (250 strings)
└─ ce.ts	# Chechen (250 strings)
└─ os.ts	# Ossetian (250 strings)

Translation API

```
// messenger/src/i18n/index.ts

// Get translation by key
export function t(key: keyof Translation, lang: Language = 'ru'): string {
  return translationsData[lang]?.[key] || translationsData.ru[key] || key;
}

// Get all translations for a language
export function getTranslations(lang: Language): Translation {
  return translationsData[lang] || translationsData.ru;
}

// Backward compatibility
export function translations(lang: Language): Translation {
  return getTranslations(lang);
}
```

Fallback Behavior

```
// Current fallback: Russian
function t(key: string, lang: Language = 'ru'): string {
  // 1. Try selected language
  if (translationsData[lang]?.[key]) {
    return translationsData[lang][key];
  }

  // 2. Fallback to Russian
  if (translationsData.ru?.[key]) {
    return translationsData.ru[key];
  }

  // 3. Return key (silent fallback)
  return key;
}
```

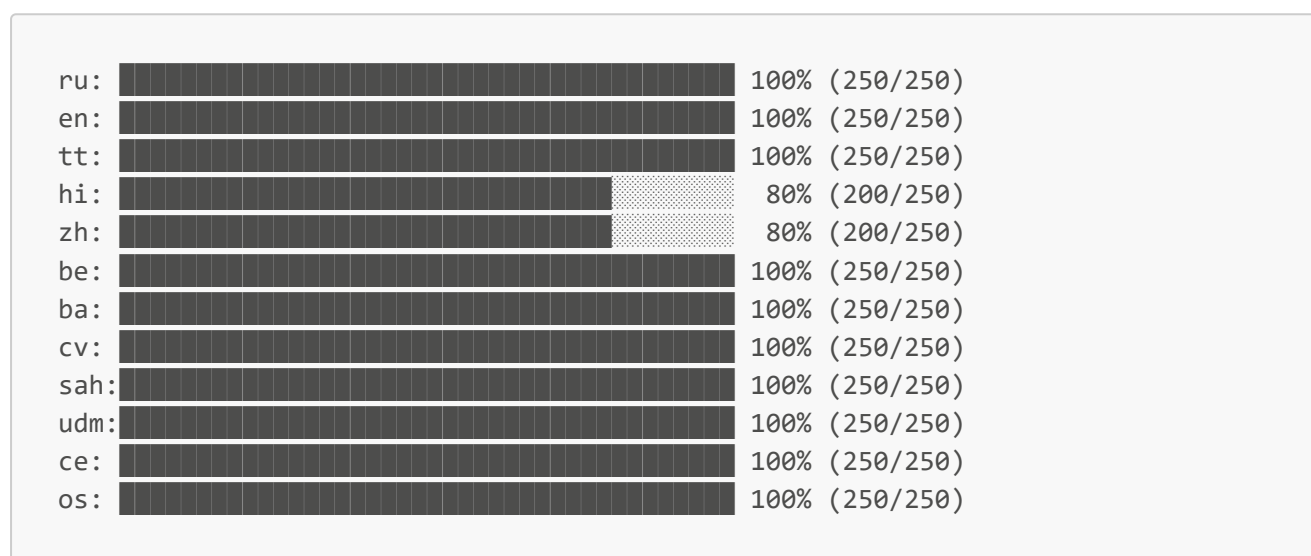
Issue: Current implementation has silent fallback. Policy requires explicit markers.

TRANSLATION COVERAGE

Overall Coverage

Metric	Value
Total Languages	12
Full Coverage (100%)	10 languages
Partial Coverage (80%)	2 languages (hi, zh)
Average Coverage	96.7%
Total Strings (ru)	250

Coverage by Language



Missing Strings (30 total)

Hindi (15 missing):

- familyRelations, addRelation, relationChildMother, relationChildFather, relationSibling, relationSpouse, selectRelation, relatedUser, noRelations, removeRelation (10 UI strings)
- installApp, installAppDesc, enableNotifications, notificationsDesc (4 PWA strings)
- contactsPermissionDenied (1 contacts string)

Chinese (15 missing):

- Same 15 keys as Hindi

☒ CONFIRMED INVARIANTS

From Code Evidence

Invariant	Evidence	Status
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Invariant	Evidence	Status
Russian as source	<code>translationsData.ru</code> is fallback	☒ Confirmed
12 languages	<code>Language</code> type in <code>types.ts</code>	☒ Confirmed
Translation API	<code>t()</code> , <code>getTranslations()</code> functions	☒ Confirmed
Fallback to ru	<code>translationsData[lang]?.[key] translationsData.ru[key]</code>	☒ Confirmed
String keys	All keys in <code>ru.ts</code>	☒ Confirmed

Policy Requirements (New)

Invariant	Status	Notes
100% translatable	 Partial	Some hardcoded strings may exist
No silent fallback	 Not implemented	Current code has silent fallback
Technical zone managed	 Future	Currently code-based
No code editing	 Future	Currently requires code changes

ARCHITECTURE ANALYSIS

Current State (Code-Based)

Adding a language requires:

1. Edit `messenger/src/i18n/types.ts` (add to `Language` type)
2. Add `messenger/src/i18n/locales/new-lang.ts`
3. Edit `messenger/src/i18n/index.ts` (import new locale)
4. Rebuild application

Future State (Technical Zone Managed)

Adding a language requires:

1. Add language metadata to `i18n-languages.json`
2. Upload translation file via UI
3. Validate coverage
4. Activate language
5. No code changes required

TRANSLATION PACKAGES

Identified Packages

Package	Description	Strings	Coverage
ui	UI components (buttons, labels, etc.)	~150	100% (ru, en), 80% (hi, zh)
system	System messages (errors, loading)	~50	100%
public	Public pages (about, downloads)	~50	100%

Package Structure

```
// Example translation keys by package
{
  // UI Package
  appName: 'Balloo',
  loading: 'Загрузка...',
  save: 'Сохранить',
  cancel: 'Отмена',

  // System Package
  error: 'Ошибка',
  success: 'Успешно',
  errorNetwork: 'Ошибка сети',

  // Public Package
  downloads: 'Загрузки',
  aboutBalloo: 'О Balloo',
  privacyPolicy: 'Положение о неразглашении'
}
```

GAPS & ISSUES

Current Issues

Issue	Severity	Status
Silent fallback	High	Needs fix
Partial coverage (hi, zh)	Medium	Needs translation
Code-based language addition	Medium	Future: technical zone
No coverage tracking	Medium	Created: i18n-missing-strings.json
No management UI	Low	Future feature

Missing Features

Feature	Priority	Notes
Explicit missing markers	High	[MISSING: key]
Coverage validation	High	Block activation if <threshold
Language management UI	Medium	Technical zone feature
Translation import/export	Medium	For translators
Audit trail	Low	Track changes

 CODEGEN RELEVANCE

For AI Code Generation

```
{
  "i18nDiscovery": {
    "sourceOfTruth": "messenger/src/i18n/",
    "languages": 12,
    "primaryLanguage": "ru",
    "fallbackLanguage": "ru",
    "coverage": {
      "average": 96.7,
      "fullCoverage": 10,
      "partialCoverage": 2
    },
  },
  "invariants": [
    "russian-as-source",
    "12-languages-baseline",
    "fallback-to-ru"
  ],
  "issues": [
    "silent-fallback",
    "partial-hi-zh-coverage",
    "code-based-addition"
  ],
  "futureState": {
    "technicalZoneManaged": true,
    "noCodeChanges": true,
    "coverageValidation": true
  }
}
```

 CREATED ARTIFACTS

Policy & Models

- ☒ `I18N_POLICY.md` — Language policy
- ☒ `I18N_LANGUAGE_MODEL.md` — Language model
- ☒ `I18N_TRANSLATION_MODEL.md` — Translation model
- ☒ `I18N_EXTENSION_MODEL.md` — Extension model
- ☒ `I18N_CODEGEN_POLICY.md` — Codegen policy
- ☒ `I18N_DOCGEN_POLICY.md` — Docgen policy
- ☒ `I18N_COVERAGE_POLICY.md` — Coverage policy
- ☒ `I18N_MANAGEMENT_MODEL.md` — Management model
- ☒ `I18N_PACKAGE_MODEL.md` — Package model
- ☒ `I18N_RUNTIME_MODEL.md` — Runtime model

State Files

- ☒ `../state/i18n-languages.json` — Language registry (12 languages)
- ☒ `../state/i18n-packages.json` — Translation packages
- ☒ `../state/i18n-translation-coverage.json` — Coverage map
- ☒ `../state/i18n-language-priority.json` — Priority map
- ☒ `../state/i18n-runtime-map.json` — Runtime mapping
- ☒ `../state/i18n-management-map.json` — Management mapping
- ☒ `../state/i18n-missing-strings.json` — Missing strings tracker

Contracts

- ☒ `../contracts/i18n/I18nPolicyContract.md` — Policy contract
- ☒ `../contracts/i18n/I18nLanguageContract.md` — Language contract
- ☒ `../contracts/i18n/I18nTranslationContract.md` — Translation contract
- ☒ `../contracts/i18n/I18nLanguageManagementContract.md` — Management contract
- ☒ `../contracts/i18n/I18nCodegenContract.md` — Codegen contract

Language Docs

- ☒ `./languages/LANGUAGE_ru.md` — Russian
- ☒ `./languages/LANGUAGE_en.md` — English
- ☒ `./languages/LANGUAGE_tt.md` — Tatar
- ☒ `./languages/LANGUAGE_hi.md` — Hindi
- ☒ `./languages/LANGUAGE_zh.md` — Chinese
- ☒ `./languages/LANGUAGE_be.md` — Belarusian
- ☒ `./languages/LANGUAGE_ba.md` — Bashkir
- ☒ `./languages/LANGUAGE_cv.md` — Chuvash
- ☒ `./languages/LANGUAGE_sah.md` — Yakut
- ☒ `./languages/LANGUAGE_udm.md` — Udmurt
- ☒ `./languages/LANGUAGE_ce.md` — Chechen
- ☒ `./languages/LANGUAGE_os.md` — Ossetian

☒ SUCCESS CRITERIA

Met ☒

- ☒ Languages reconstructed from messenger
- ☒ 12 languages documented
- ☒ Coverage tracked (96.7% average)
- ☒ Missing strings identified (30 total)
- ☒ Policy documented
- ☒ State files created
- ☒ Contracts created
- ☒ Language docs created
- ☒ Future architecture defined

Pending 

-  Implement explicit missing markers in code
-  Complete hi, zh translations
-  Build language management UI
-  Integrate with technical zone

RELATED DOCUMENTS

- [I18N_INDEX.md](#) — I18N documentation index
- [I18N_POLICY.md](#) — Language policy
- [../state/i18n-languages.json](#) — Language registry
- [../state/i18n-missing-strings.json](#) — Missing strings

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 **Baloo - Проверни общение!**